

# Mohd Shayan

New Delhi, India | +91 8920-038741 | [contact@mohdshayan.com](mailto:contact@mohdshayan.com) | [mohdshayan.com](http://mohdshayan.com)  
[linkedin.com/in/shayanmohd](https://linkedin.com/in/shayanmohd) | [github.com/shayanmohd](https://github.com/shayanmohd)

## PROFESSIONAL SUMMARY

**Software Engineer and Founder** who owns products end to end: full-stack architecture, cloud and DevOps infrastructure, applied AI, and on-chain systems. Founder of **SocialSure Private Limited**, an AI-first company, and of its software services brand **CustomGlide**, delivering enterprise CRM, QR ticketing, and automation platforms to clients across **Malaysia, Canada, and the UAE**. Ships at volume: **eight Android applications live on Google Play**, a real-time cross-exchange Web3 arbitrage terminal with MEV-protected atomic execution, and a fine-tuned LLM helpdesk assistant for India's national railway IT authority.

Author of **computational linguistics research selected for the Annual Meeting of the Association for Computational Linguistics (ACL)**. Deep specialisation across the computer science stack, spanning distributed systems, applied cryptography, and low-level mobile engineering through to LLM fine-tuning and smart contract security, alongside sustained philanthropic work funding education, nutrition, and digital access for children in need. Elected **Fellow of the Royal Society of Arts (FRSA)** and **Fellow of the Royal Geographical Society (FRGS)**.

## EXPERIENCE

### SocialSure Private Limited

New Delhi, India / Remote

*Founder & Chief Executive Officer*

*March 2024 – Present*

- Founded and lead an **AI-first product company**, owning product vision, engineering, and commercial strategy across AI training and data pipelines, an AI-native social media ecosystem, and a portfolio of applied-AI products.
- Architected and deployed the core **SaaS platform** as a **Turborepo** monorepo in **Next.js** and **TypeScript**, with backend services in **Node.js**, **Express**, and **PostgreSQL** (Prisma / Drizzle) enforcing schema migrations and transactional consistency.
- Lead **Android** development end to end, publishing **eight applications to the Google Play Store** under the company developer account, including SocialSure's own native app, and owning release engineering, signing, policy compliance, and staged rollouts.
- Built **CI/CD** pipelines and **Docker**-containerised deployments with structured logging, metrics, and alerting, giving a small team enterprise-grade release safety and high availability under production load.

### CustomGlide (a SocialSure brand)

Remote / International

*Founder & Lead Software Engineer*

*2024 – Present*

- Delivered contract software for enterprise clients across **Malaysia, Canada, and the UAE**, owning the full cycle from scoping and architecture through delivery, handover, and post-launch support.
- Built a unified **CRM platform** covering visual sales pipelines, marketing campaign execution, and service ticketing with real-time dashboards, designed as a migration target for teams leaving Salesforce, HubSpot, and Zoho ([demo.customglide.org](http://demo.customglide.org)).
- Engineered a **QR ticketing and contactless entry system** for visitor attractions: online booking, gateless scan-in validation, anti-replay ticket verification, and live visitor analytics ([demoqr.customglide.org](http://demoqr.customglide.org)).
- Ship **AI automation**, workflow integration, and security-hardened cloud deployments as a retained engineering service across concurrent client engagements.

### Centre for Railway Information Systems (CRIS)

New Delhi, India

*Machine Learning Intern (Practical Trainee)*

*July 2025 – Sep 2025*

- Engineered a conversational **AI chatbot** for the EPS helpdesk by fine-tuning open-source **LLMs** (Llama, Mistral) with **Hugging Face Transformers** and parameter-efficient tuning.
- Architected a **RAG (Retrieval-Augmented Generation)** pipeline using **LangChain** and vector databases, with context-aware retrieval and grounded citation to cut hallucination on domain-specific queries.
- Optimised inference latency and served the system on **Oracle Cloud** in **Python**, tuning batching, quantisation, and retrieval depth against production response-time targets.

## PUBLICATIONS & RESEARCH

**Implementation and Analysis of the CYK Algorithm for Context-Free Grammar Recognition: Algorithmic Complexity, Probabilistic Extensions, and Theoretical Boundaries.** Selected for the Annual Meeting of the Association for Computational Linguistics (ACL). B.Tech dissertation, Department of Applied Mathematics, Delhi Technological University, supervised by Dr. Laxminarayan Das, May 2026.

Derived a deterministic CYK parser from first principles with proved  $O(n^3 \cdot |G|)$  time and  $O(n^2)$  space bounds, extended it to **Probabilistic CFGs** with **Viterbi decoding** in the log-probability domain for ambiguity resolution, and applied the **Pumping Lemma** to establish the formal limit of context-free recognition. Benchmarked against LR(1), Earley, and recursive-descent parsers and related to modern **Compound PCFG** and Transformer-constrained neural parsing.

## PROJECTS

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### Shayanomaly (Web3 Arbitrage & Trading Terminal) | *Next.js, Node.js, Solidity, CCXT, Flashbots*

- Built a browser-based quantitative **trading terminal** streaming live order books from five CEX and DEX venues over **WebSockets** via **CCXT**, rendered in **Next.js** and **Tailwind CSS**.
- Engineered a **Node.js** execution engine detecting cross-venue arbitrage anomalies and executing **Aave V3 flash loans** through custom **Solidity** contracts as a single atomic transaction.
- Implemented MEV protection with the **Flashbots Bundle Provider** to bypass the public mempool, giving front-running-resistant execution with zero private key exposure.

### Habits: Minimalist Productivity Tracker (Android) | *Kotlin, Jetpack Compose, MVVM, Room, CI/CD*

- Shipped a privacy-first, fully offline Android app to the **Google Play Store**, built entirely in **Jetpack Compose** with **Material 3**, a **Glance** home-screen widget, and nine-language localisation.
- Architected an **MVVM** system on **Room**, **Kotlin Coroutines**, and **StateFlow** for zero-latency, 100% on-device storage, with a **GitHub Actions** pipeline automating build verification, Gradle caching, and zero-leak signing-key handling.

### Turup Chaal (Multiplayer Card Game) | *Unity, C#, Photon PUN2, Authoritative Netcode*

- Built a production Court Piece trick-taking game for Android solo in **Unity** and **C#**, with a pure-C# rules engine decoupled from the render layer, and host-authoritative **Photon PUN2** multiplayer using per-player state redaction to prevent hand-inspection cheating by construction.

### Selected Additional Work | *React Native, Python, Java, Cryptography*

- **deAIfy**: cross-platform **React Native / Expo** tool stripping AI writing tells entirely on-device, with optional LLM rewriting via a user-supplied key so no text leaves the client by default.
- **ShreezyPay Bot**: Telegram escrow-mediation bot running unlimited concurrent group pairs with per-user send permissions, keyword filtering, and reply-thread sync preserving counterparty anonymity.
- **Minimalist Calculator**: native Android calculator concealing an **AES-GCM** encrypted media vault keyed through the **Android Keystore**, unlocked by a secret tap sequence.

## TECHNICAL SKILLS

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**Languages**: TypeScript, JavaScript (Node / Bun / Browser), Python, Rust, C++, Kotlin, Java, C#, Solidity, SQL, Bash, HTML/CSS

**Frontend**: React, Next.js, React Native, Expo, Tailwind CSS, Vue.js, Three.js, Jetpack Compose, Material 3, Responsive and Accessible UI

**Backend & APIs**: Node.js, Express, REST, WebSockets, WebRTC, Redis Pub/Sub, Event-Driven Architecture, Microservices, Auth (OAuth, JWT, RBAC), Rate Limiting, Caching

**Databases**: PostgreSQL, MongoDB, SQLite / Room, Redis, Vector Databases, Prisma, Drizzle, Schema Design, Query Optimisation, Migrations

**DevOps & Cloud**: Docker, Kubernetes, AWS (EC2, S3, ASGs), Oracle Cloud, Terraform (IaC), CI/CD, GitHub Actions, Cloudflare Workers, CDNs, Reverse Proxies, Firecracker, Linux, Observability and Monitoring

**AI / ML**: LLM Fine-Tuning (LoRA / PEFT), RAG Pipelines, LangChain, Hugging Face Transformers, PyTorch, TensorFlow, Scikit-learn, Agent Frameworks, RLHF, Neural Networks (CNN / RNN / LSTM), Embeddings and Semantic Search, Prompt Engineering, Model Evaluation

**Web3 & Blockchain**: Solana Architecture, Anchor, Rust Smart Contracts, @solana/web3.js, Ethereum, Solidity, Ethers.js, Hardhat, Foundry, IPFS, DeFi (AMM, CLMM, Flash Loans), Token Programs, MEV Protection / Flashbots, Contract Security

**Security & Cryptography**: Applied Cryptography, AES-GCM, MPC, Shamir's Secret Sharing, Key Management, Hardware-Backed Keystores, Secure SDLC, Secrets Handling

**Mobile & Games**: Android (Kotlin, Java), MVVM, Coroutines / StateFlow, Room, Glance Widgets, Play Store Release Engineering, Unity, Photon PUN2, Authoritative Netcode

**Foundations**: Data Structures and Algorithms, System Design, Distributed Systems, Concurrency, Applied Mathematics (Linear Algebra, Probability, Optimisation), Git, Testing and Code Review, Technical Leadership

## EDUCATION

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### Delhi Technological University (formerly DCE)

*B.Tech in Mathematics & Computing*

New Delhi, India

Nov 2022 – May 2026

### Delhi Public School, Mathura Road

*Complete Schooling to Senior Secondary Certificate (Class XII), CBSE*

New Delhi, India

2007 – 2021

## HONOURS, FELLOWSHIPS & CERTIFICATIONS

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**Fellow of the Royal Society of Arts (FRSA)**, London, United Kingdom

**Fellow of the Royal Geographical Society (FRGS)**, London, United Kingdom

**100xSchool Combined Bootcamp** (Full-Stack Web Development, DevOps, Machine Learning & AI, Web3, DSA), 100xDevs / Harkirat Singh, 2026

**100 Days of Code: The Complete Python Pro Bootcamp**, Udemy, June 2025 | ID: UC-bee87944-97c0-4e65-8f0b-57c098ed209e

**Supervised Machine Learning: Regression and Classification**, DeepLearning.AI, Mar 2025 | ID: K6PL95XZFD76

**Introduction to Self-Determination Theory**, University of Rochester (Coursera), May 2022 | ID: G7UDMQN9WTWR